Md Ashik Khan

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Research Interests

Computer Vision, Medical Image Analysis, Multimodal Deep Learning, Video Action Recognition, Trustworthy AI

Education

Indian Institute of Technology Kharagpur — West Bengal, India
Master of Technology in Computer Science and Engineering — *2021 – 2023*

- CGPA: 8.76/10.00 • Advisor: Dr. Abir Das • Scholarship: ICCR (fully funded)
- Thesis: *A Large-scale Study of Representation Learning and the Benchmarking of Video Action Recognition*

Bangladesh Army University of Science and Technology — Saidpur, Bangladesh
Bachelor of Science in Computer Science and Engineering — *2015 – 2019*

- CGPA: 3.35/4.00 (Final-year GPA: 3.61/4.00)

Publications

Refereed Publications

1. **Md Ashik Khan**, Abu Saleh Musa Miah, Fahmid Al Farid, Md Abdur Rahim, and Hezerul Abdul Karim. "Human Activity Recognition Under Low-Light Conditions Using Frozen CLIP with Lightweight Adaptation." *Discover AI* (Springer Nature), 2026 (Accepted).
2. **Md Ashik Khan** and Md Nahid Siddique. "SHIFT-M3: Pre-fusion Alignment-based Consistency Screening for Multimodal ECG Record Integrity." *Machine Learning for Healthcare (MLHC 2026)* (Accepted).
3. **Md Ashik Khan** and Md Nahid Siddique. "Fixed-Budget Parameter-Efficient Training with Frozen Encoders Improves Multimodal Chest X-Ray Classification." *Proceedings of the 28th International Conference on Computer and Information Technology (ICCIT 2025)*, Cox's Bazar, Bangladesh, December 2025, pp. 2748-2753. [DOI].
4. **Md Ashik Khan** and Arafat Alam Jion. "Fixed-Threshold Evaluation of a Hybrid CNN-ViT for AI-Generated Image Detection Across Photos and Art." *Proceedings of the 28th International Conference on Computer and Information Technology (ICCIT 2025)*, Cox's Bazar, Bangladesh, December 2025, pp. 3853-3858. [DOI].
5. **Md Ashik Khan** and Rafath Bin Zafar Auvee. "Comparative Analysis of Resource-Efficient CNN Architectures for Brain Tumor Classification." *Proceedings of the 27th International Conference on Computer and Information Technology (ICCIT 2024)*, Cox's Bazar, Bangladesh, December 2024, pp. 639-644. [DOI].

Under Review

- Md Nahid Siddique[*], **Md Ashik Khan**[*], Md Rezaul Karim Khan, Naphtali David Rishe, and Dongsheng Luo. "High Accuracy, Weak Transfer: A Cross-Domain Audit of Fake News Detection." Submitted to *WI-IAT 2026*. (*Equal contribution)
- **Md Ashik Khan**, Abu Saleh Musa Miah, Md Nahid Siddique, and Md Abdur Rahim. "Prompt Sensitivity and Answer-Position Bias in VideoLLM Action Recognition." Submitted to *WACV 2027*.

Ongoing Research

- **Md Ashik Khan**, Abu Saleh Musa Miah, Md Abdur Rahim, and Fahmid Al Farid. "Reliability-Aware Cross-Modal Consistency for Label-Efficient Human Activity Recognition."
- **Md Ashik Khan**, Abu Saleh Musa Miah, and Md Abdur Rahim. "WICA-Net: A Lightweight Wavelet-Integrated Coordinate-Attention Network for Generalizable Brain Tumor MRI Analysis."

Research Experience

CVIR Lab, IIT Kharagpur — West Bengal, India
Graduate Research Assistant (M.Tech Thesis Research) | Advisor: Dr. Abir Das — *Jun 2022 – Apr 2023*

- **Thesis:** *A Large-scale Study of Representation Learning and the Benchmarking of Video Action Recognition*
- Systematically evaluated transferability across 6 SOTA video architectures (SlowOnly, TimeSformer, SIFAR) on 14 downstream benchmarks for self-supervised and supervised representations across 2D/3D CNNs and Video Transformers.

- Created and validated the Construction Action Dataset (1,844 annotated videos across 13 industrial activity categories) to address domain-specific data scarcity in action recognition.
- Mapped optimal transfer pathways under domain-shift and data-scarcity constraints through benchmarking out-of-domain generalization and representation alignment.

Independent Research Collaborations *May 2023 – Present*

Collaborators: Dr. Abu Saleh Musa Miah (Dept. of CSE, University of Rajshahi, Bangladesh) · Dr. Fahmid Al Farid (Faculty of CS & Informatics, BSBI, Germany) · Dr. Md Abdur Rahim (Dept. of CSE, PUST, Bangladesh) · Dr. Hezerul Abdul Karim (FAIE, Multimedia University, Malaysia)

- Designed a parameter-efficient temporal adaptation framework for frozen CLIP, achieving a +6.01% accuracy gain under low-light domain shifts with under 0.2% additional computational overhead. *(Discover AI, Springer Nature, 2026)*
- Evaluated PEFT adapter variants under a fixed parameter budget on frozen multimodal encoders, outperforming full fine-tuning by 13.8 pp AUROC (0.908 vs. 0.770) with 40× fewer parameters. *(ICCIT 2025)*
- Designed SHIFT-M3, a pre-fusion alignment-based screening method to ensure multimodal ECG record integrity and flag data misalignment under noisy clinical collection conditions. *(MLHC 2026, accepted)*
- Designed WICA-Net by integrating wavelet decomposition with coordinate attention to capture multi-scale frequency features for generalizable brain tumor MRI analysis. *(in preparation)*
- Developed RACM, a semi-supervised skeleton-IMU HAR framework using inter-branch Jensen-Shannon agreement as a training signal, raising missing-skeleton accuracy from 23.6% to 45.2% (+21.6 pp) and exceeding all baselines by >10 pp on both missing-modality metrics under 25% labeled data. *(in preparation)*
- Conducted a controlled response-regime audit of 5 VideoLLMs across 8 action-recognition datasets under 3 interfaces, uncovering 23–96 pp accuracy swings on identical clips from prompt changes alone. *(submitted, WACV 2027)*

Selected Projects

Occlusion-Robust Pedestrian Detection with YOLOv12-S and Transformer Backbones *Autumn 2025*

- Benchmarked YOLOv12-S (9.25M parameters, 21.5 GFLOPs) with Swin Transformer and ConvNeXt backbones on EuroCity, comparing global context modeling against local feature extraction for pedestrian detection.
- Designed a stratified occlusion robustness evaluation across four levels (none, light, moderate, heavy) using occlusion-aware dataset variants with per-image metadata.
- Achieved mAP50 ≈ 75% and precision ≈ 92% overall; heavy-occlusion mAP50 of 75.3% demonstrated Swin Transformer backbone resilience under severe partial visibility.

Deep Learning-based Hidden Camera Generalization Study *Spring 2024*

- Synthesized a domain-randomized dataset of surveillance camera placements to address real-world data scarcity in covert camera detection.
- Fine-tuned YOLOv8 and ResNet50 with custom augmentation pipelines to analyze generalization under heavy occlusion and low-resolution conditions, achieving 92% mAP.

Context-Specific Quote Recommendation from Historical Text *Autumn 2022*

IIT Kharagpur

- Built a context-based quote recommendation system using Hugging Face DistilBERT, applying full fine-tuning on the Quotation POTUS dataset for domain-specific semantic text retrieval.
- Achieved 82.25% accuracy on quote phrase matching, validating the effectiveness of transformer-based transfer learning on domain-specific historical text.

Professional Experience

Auptimate Dhaka, Bangladesh

Senior Software Engineer *Jan 2026 – Present*

Software Engineer *May 2023 – Dec 2025*

- Architected a scalable e-signature and document automation platform with microservices (FastAPI, React.js, PostgreSQL, AWS), supporting end-to-end digital onboarding for enterprise clients.
- Engineered an NLP-powered AI agent for automated legal document processing and semantic query resolution.

Hovata Technologies Dhaka, Bangladesh

Assistant Programmer *Mar 2020 – Aug 2021*

- Built and maintained PetrolERP for petroleum business management (sales, stock, expenses, vendors), implementing real-time inventory tracking that reduced stock discrepancies by 35% via transaction locks.

- Led development of parking management system optimizing space utilization using React.js, Redux, PHP Lumen, and MySQL.

Teaching Experience

Graduate Teaching Assistant | **Department of CSE, IIT Kharagpur**

- **Software Engineering** *Spring 2022*
- **Algorithms-I** *Autumn 2022*

Responsibilities: Conducted tutorials, evaluated assignments, mentored 20 students during laboratory sessions, and supervised multiple groups during semester projects.

Academic Service & Certifications

- **Journal Reviewer:** *IEEE Access*
- **Fundamentals of Accelerated Data Science** • NVIDIA Deep Learning Institute *2021*

Scholarships & Awards

- **ICCR Scholarship for M.Tech at IIT Kharagpur** • Fully Funded international scholarship *2021 – 2023*
- **Finalist** • BAUST-IEEE Idea Contest • BAUST *2018*
- **Honorable Mention** • Inter-University Programming Contest • IUBAT, Dhaka *2016*
- **Board Talent Rank 35th** • Secondary School Certificate (SSC) Examination, Dhaka Board *2012*

Technical Skills

Languages: Python, C++, SQL
AI & Vision: PyTorch, OpenCV, MMAction2, YOLO
Tools: Git, Linux, LaTeX, AWS